AGENTIC AI GOVERNANCE
TRUST | ASSURE | ENABLE

How Do We Prove **Transition Integrity** in Agentic AI?

Not T_4. Not T_5. The Space Between Them.

A trustworthy AI decision does not automatically prove trustworthy execution.

The assurance gap is addressed by the
Agentic AI $T_4 \rightarrow T_5$ Assurance Cone Framework™ V3.0
Transition Integrity | Execution-Path Coverage | Boundary Integrity | Independent Assurance

SIX ASSURANCE FUNCTIONS. ONE ACCOUNTABLE CONCLUSION. (I-A-N-E-T-F)

1 INTENT
What was intended and why does it matter?
- Consequential objective
- Affected resource
- Materiality
- Permitted / prohibited purpose
- Applicable constraints

2 AUTHORITY
Is the transition authorized?
- Who/what can authorize
- Scope and duration
- Resource and action
- Authority state at execution point
- Delegations and constraints

3 ENFORCEMENT
What actually controlled execution?
- Policy vs actual enforcement
- Permit / deny / contain / terminate
- Control effectiveness
- No bypass paths
- Enforcement at commit point

4 TESTING
Can the conclusion be independently challenged?
- Adversarial testing
- Stale or revoked authority
- Parallel / asynchronous paths
- Replay / duplication / bypass
- Execution-path coverage
- Failure and recovery scenarios

5 TRACEABILITY & EVIDENCE
Can the transition be reconstructed?
- Who → What → Why
- Authority → Policy → Path
- Enforcement → Commit™
- Downstream lineage
- Evidence integrity and provenance

6 FORENSICS & REVOCABILITY
Can the transition be traced and contained?
- Event reconstruction
- Exposure and containment
- Revocation effectiveness
- Residual effects
- Accountability and responsibility

ENTERPRISE CONTEXT
RISK | PEOPLE | PROCESSES | TECHNOLOGY | DATA | VALUE

THE $T_4 \rightarrow T_5$ ASSURANCE BOUNDARY (V3.0)

Must be established, within the defined assurance scope:

AUTHORITY VALIDATION
Was the relevant authority valid at the applicable commit point?

TRANSITION CONTROL
Were required controls active and effective?

EXECUTION-PATH COVERAGE
Were all consequential execution paths identified, controlled, and accounted for?

INTERVENTION / REVOCATION EFFECTIVENESS
If intervention occurred, did it become operationally effective?

EVIDENCE RECONCILIATION
Do independent evidence sources establish a consistent transition sequence?

BOUNDARY INTEGRITY
Does the assurance conclusion remain within the demonstrated evidence and execution boundary?

NO ASSURANCE CLAIM MAY EXTEND BEYOND WHAT CAN BE INDEPENDENTLY DEMONSTRATED.

KEY V3.0 PRINCIPLES

- AUTHORITY ≠ CAPABILITY
- AUTHORIZATION ≠ EXECUTION
- EVIDENCE VALIDITY ≠ EVIDENCE COMPLETENESS
- COVERAGE MUST BE DEMONSTRATED
- UNKNOWN COVERAGE IS A LIMITATION
- NO CAPTURED ACT ≠ NO ACT OCCURRED
- ASSURANCE ≠ ASSERTION
- CONCLUSION CANNOT EXCEED EVIDENCE
- ALIGNMENT ≠ EXTERNAL CONFORMITY
- FRAMEWORK USE ≠ CERTIFICATION

THE GOVERNING ASSURANCE QUESTION (V3.0)

Can an independent examiner establish — within the defined assurance scope — that the consequential transition from T_4 to T_5 occurred through an authorized and controlled execution path, and that no unaccounted consequential execution path exists capable of producing the relevant effect within that scope?

YES	WITH LIMITATIONS	NO
Transition Integrity Demonstrated	Partially Demonstrated	Transition Integrity Unproven
Accept the conclusion	Qualify the conclusion	Reject the conclusion

THE $T_4 \rightarrow T_5$ LIFECYCLE (V3.0)

T_0	T_1	T_2	T_3	T_4		T_5	
Authority Change	Propagation	Observation	Consequential Attempt	Commit-Time Seal	Act-Specific Evidence	Authoritative Enforcement	Downstream Inheritance

Adnan Shamim
Founder | The First Agentic AI Audit Ecosystem™

RESPONSIBLE AI | INDEPENDENT ASSURANCE | REAL-WORLD IMPACT

TRUST
BUILDS
TOMORROW

THE AGENTIC AI $T_4 \rightarrow T_5$ ASSURANCE CONE FRAMEWORK™ V3.0
Independent Assurance for a Safer, More Accountable AI Future.

THE AGENTIC AI $T_4 \rightarrow T_5$ ASSURANCE CONE FRAMEWORK™

V3.0 — Enterprise Assurance Edition

Transition Integrity • Execution-Path Coverage • Boundary Integrity • Independent Assurance

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1. PURPOSE

The Agentic AI $T_4 \rightarrow T_5$ Assurance Cone Framework™ establishes a technology-neutral independent-assurance structure for evaluating whether a consequential transition from an observed lifecycle state through commit-time execution to an authoritative enforcement and downstream state is sufficiently governed, attributable, controlled, evidenced, reconstructable, and bounded.

The Framework is independently structured and does not prescribe a particular technology, architecture, vendor, protocol, implementation pattern, or control product.

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2. FRAMEWORK STATUS & NON-SUBSTITUTION

The Agentic AI $T_4 \rightarrow T_5$ Assurance Cone Framework™ is an independently developed assurance framework.

It is not intended to:

- replace applicable laws or regulations;
- replace organizational governance requirements;
- replace applicable risk-management requirements;
- replace information-security requirements;
- replace internal-control requirements;
- replace internal-audit professional requirements;
- establish regulatory approval;
- establish certification or accreditation;
- automatically establish compliance with any external requirement.

The Framework may be mapped to, used alongside, or incorporated within an organization's existing governance, risk, compliance, security, audit, and assurance architecture.

Application of this Framework does not, by itself, establish conformity with any external requirement.

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3. EXTERNAL REQUIREMENTS & ALIGNMENT

Where an organization is subject to external laws, regulations, contractual obligations, professional requirements, governance requirements, or other control obligations, those requirements remain independently applicable.

V3.0 may be used as an assurance overlay to evaluate the $T_4 \rightarrow T_5$ transition against those applicable requirements.

The organization must separately determine:

- which external requirements apply;
- which requirements are mandatory;
- which controls are required;
- which evidence is required;
- which conformity criteria apply;
- which reporting obligations exist;
- which regulatory or contractual consequences apply.

Governing Principle

Framework alignment does not establish external conformity.

Therefore:

V3.0 Application ≠ External Compliance

Framework Alignment ≠ Certification

Assurance Conclusion ≠ Regulatory Approval

Control Evidence ≠ Automatic Compliance

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4. CLAIMS DISCIPLINE

Any assurance statement produced using V3.0 must clearly identify, where material:

1. the assurance population;
2. the relevant time period;
3. the execution-path boundary;
4. the evidence boundary;
5. material exclusions;
6. material limitations;
7. the assurance classification;
8. the applicable control requirements;
9. the basis for the conclusion.

No statement may imply a broader assurance conclusion than the underlying evidence, coverage, authority, enforcement, or defined boundary supports.

No V3.0 conclusion may be represented as certification, accreditation, regulatory approval, or automatic conformity with an external requirement.

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5. FUNDAMENTAL ASSURANCE PROBLEM

A trustworthy AI decision does not automatically establish trustworthy execution.

Likewise:

Observed State ≠ Authoritative State

Capability ≠ Authority

Authority ≠ Authorization

Authorization ≠ Execution

Execution ≠ Effect

Evidence Validity ≠ Evidence Completeness

Captured Execution ≠ Complete Execution Population

Propagation Completion ≠ Retroactive Authorization

T_5 Enforcement ≠ Retroactive Validation of T_4

Therefore, trustworthy endpoints alone cannot establish transition integrity.

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6. THE $T_4 \rightarrow T_5$ ASSURANCE SEAM

T_4 — OBSERVATION & COMMIT-TIME SEALING

T_4 establishes what the actor observed and what transition context was sealed, including where applicable:

- lifecycle state;
- freshness;
- propagation context;
- execution path;
- attempted transition;
- relevant conditions;
- commit-time context.

T_4 observation does not, by itself, establish authoritative authorization.

ACT-SPECIFIC COMMIT EVIDENCE

A consequential act requires evidence capable of binding:

ACT + ACTOR + AUTHORITY + POLICY + PATH + ENFORCEMENT + TIME + COMMIT + EFFECT + LINEAGE

T_5 — AUTHORITATIVE ENFORCEMENT & DOWNSTREAM INHERITANCE

T_5 establishes what the authoritative enforcement environment actually permitted, denied, constrained, contained, terminated, or otherwise governed.

T_5 must not be used as retroactive authorization for an earlier consequential act.

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7. T_0–T_5 ADVERSARIAL TRACE

T_0 — AUTHORITY CHANGE

Revocation, suspension, restriction, expiry, policy change, or other material authority event.

T_1 — PROPAGATION

Changed authority state propagates through relevant systems, agents, services, queues, credentials, caches, delegated processes, and execution paths.

T_2 — OBSERVATION

Actor/agent observes the available lifecycle or authority state.

T_3 — CONSEQUENTIAL ATTEMPT

Actor/agent initiates or attempts a consequential action.

T_4 — COMMIT-TIME OBSERVATION & SEAL

Observed state, freshness, execution path, and transition context are preserved.

ACT-SPECIFIC EVIDENCE

Evidence binds the consequential act to the applicable authority and enforcement conditions.

T_5 — AUTHORITATIVE ENFORCEMENT

Authoritative control determines the downstream treatment of the transition.

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8. I-A-N-E-T-F ASSURANCE CONE™

I — INTENT

Establish:

- intended objective;
- consequential action;
- affected resource;
- materiality;
- permitted purpose;
- prohibited purpose;
- applicable constraints.

A — AUTHORITY

Establish:

- who/what could authorize;
- scope;
- duration;
- resource;
- action;
- applicable conditions;
- authority state at the relevant execution boundary.

N — ENFORCEMENT

Establish what actually controlled execution.

Distinguish:

Policy Exists

from

Policy Was Enforced

from

Enforcement Affected Execution

E — TESTING

Test whether the assurance claim survives adversarial conditions, including:

- revoked authority;
- stale authority;
- propagation delay;
- policy drift;
- failed enforcement;
- bypass;
- replay;
- duplication;
- parallel execution;

- asynchronous execution;
- delegated execution;
- cached authorization;
- alternate credentials;
- failover;
- recovery;
- compromised evidence generation.

Execution-Path Coverage is a mandatory Testing dimension.

T — TRACEABILITY & EVIDENCE

Establish:

Who → What → Why → Authority → Policy → Path → Enforcement → Commit → Effect → Downstream Lineage → Evidence

F — FORENSICS & REVOCABILITY

Establish the ability to:

- reconstruct events;
- identify authority;
- identify execution path;
- establish exposure;
- establish containment;
- demonstrate revocation effectiveness;
- identify residual effects;
- preserve evidence;
- determine responsibility;
- identify where the assurance boundary was exceeded.

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9. EXECUTION-PATH COVERAGE

Execution-Path Coverage is a mandatory assurance condition.

The claimed consequential execution population must be defined.

Potential paths include:

- primary execution;
- parallel execution;
- asynchronous queues;
- scheduled execution;
- delegated agents;
- downstream agents;
- retries;
- cached authorization;
- pre-authorized workflows;
- long-running processes;
- service-to-service execution;
- alternate credentials;
- failover;
- recovery;
- emergency mechanisms;
- administrative paths;
- external integrations.

A path capable of producing the claimed consequence cannot be silently assumed to be covered.

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10. DEMONSTRABLE COVERAGE CRITERIA

Execution-path coverage is **Demonstrated** only where applicable evidence establishes:

C1 — POPULATION DEFINITION

The consequential execution population is explicitly identified.

C2 — PATH IDENTIFICATION

Known paths capable of producing the consequence are identified.

C3 — CONTROL BOUNDARY

Each claimed covered path is subject to the required authority and enforcement boundary.

C4 — OBSERVABILITY

Relevant consequential activity is sufficiently observable to support the specific assurance claim.

C5 — EVIDENCE BINDING

Consequential acts are attributable to the relevant execution path and commit context.

C6 — BYPASS ASSESSMENT

Known alternate, parallel, asynchronous, delegated, recovery, and failover paths are addressed.

C7 — EXCLUSION DISCLOSURE

Any path outside coverage is explicitly identified.

C8 — INDEPENDENT RECONSTRUCTION

An examiner can independently reconstruct the relevant transition and determine the applicable assurance boundary.

C9 — FAILURE CONSEQUENCE

The predetermined assurance consequence of insufficient coverage is defined.

Coverage is not demonstrated merely because the primary path is instrumented.

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11. COVERAGE CLASSIFICATION

Every relevant execution path should be classified as:

COVERED

Required authority, enforcement, observability, and evidence conditions are demonstrated.

CONDITIONALLY COVERED

Coverage exists only under explicitly defined conditions.

UNVERIFIED

Coverage cannot be independently demonstrated.

EXCLUDED

The path is deliberately outside the assurance boundary.

FAILED

The path was demonstrated to violate a required assurance condition.

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12. EXECUTION-PATH COVERAGE INVARIANT

An assurance conclusion shall not extend beyond the consequential execution population and execution paths for which the required authorization, enforcement, detection, and evidence boundary can be independently demonstrated.

Therefore:

Covered → Assurance Permitted

Conditionally Covered → Assurance Qualified

Unverified → Assurance Limited

Excluded → Outside Conclusion

Failed → Control/Assurance Failure

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13. BOUNDARY INTEGRITY

The assurance boundary must explicitly state:

- what is included;
- what is excluded;
- what is controlled;
- what is observable;
- what is evidenced;
- what is reconstructable;
- what remains unknown;
- what consequence follows from the limitation.

Boundary Integrity Rule

Unknown coverage is an assurance condition—not evidence of absence.

Therefore:

No Captured Act ≠ No Act Occurred

No Observed Violation ≠ Demonstrated Prevention

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14. PROPAGATION-WINDOW ASSURANCE

Distributed-system propagation latency is not automatically a control failure.

However, consequential exposure during the propagation window requires predefined treatment.

The assurance boundary should establish, where applicable:

- maximum exposure duration;
- affected authority scope;
- permitted actions;
- prohibited actions;
- enforcement controls;
- containment;
- compensating controls;
- monitoring;
- escalation;
- accountability;
- residual-risk ownership;
- boundary-exceedance criteria;
- required evidence.

Critical distinction

Measuring Exposure ≠ Proving Prevention

15. MULTI-AGENT & DELEGATED AUTHORITY

Where authority moves between agents, systems, or services, sufficient lineage must establish:

Origin → Delegation → Authority → Policy → Execution Path → Enforcement → Commit → Effect → Downstream Inheritance

Delegation must not silently expand effective authority.

16. ALTERNATE-PATH ASSURANCE

Any alternate path capable of producing the claimed consequence must be:

Controlled

or

Evidenced

or

Detected

or

Explicitly Excluded

or

Treated as an Assurance Limitation

17. ACT-SPECIFIC EVIDENCE REQUIREMENT

A consequential act should have evidence capable of binding:

ACT + ACTOR + AUTHORITY + POLICY + PATH + ENFORCEMENT + TIME + COMMIT + EFFECT + LINEAGE

Evidence should be:

- attributable;
- act-specific;
- time-relevant;

- authority-bound;
- context-bound;
- tamper-evident;
- replay-resistant;
- substitution-resistant;
- independently verifiable;
- reconstructable.

18. NON-EXECUTION ASSURANCE RULE

V3.0 does not require an impossible universal negative over an uncontrolled distributed environment.

Instead:

A non-execution conclusion is valid only within the demonstrated consequential execution population and execution-path boundary.

Where execution-path coverage is incomplete:

Non-execution is not demonstrated beyond the verified boundary.

Therefore:

No Evidence of Execution ≠ Evidence of No Execution

unless the relevant execution population and capable paths are demonstrably covered by the applicable control and evidence boundary.

19. FALSIFICATION CONDITIONS

The assurance claim is falsified, limited, or failed where testing demonstrates:

1. authority cannot be established;
2. enforcement cannot be established;
3. T_4 observation is treated as authoritative without supporting evidence;
4. T_5 is used to retroactively justify T_4;

5. act-specific evidence is absent;
6. evidence is not uniquely bound to the act;
7. evidence is replayable, substitutable, or alterable;
8. an alternate path bypasses the boundary;
9. execution-path coverage cannot be demonstrated;
10. consequential execution population is undefined;
11. delegated authority expands without authorization;
12. propagation exposure exceeds its predefined boundary;
13. evidence relies solely upon retrospective assertion;
14. independent reconstruction is impossible;
15. assurance conclusion exceeds demonstrated coverage;
16. a consequential path exists outside the claimed assurance boundary without explicit treatment.

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20. INDEPENDENT RECONSTRUCTION TEST

An independent examiner must be able to determine:

Authority — What authority existed?

Observation — What did the actor observe?

Governing State — What authority/policy governed the act?

Enforcement — What actually controlled execution?

Path — Through which path did execution occur?

Coverage — Was that path demonstrably covered?

Commit — When did the consequential act commit?

Effect — What consequence occurred?

Lineage — What downstream actions inherited authority?

Evidence — Can the evidence be independently verified?

Limitation — What remains outside the assurance boundary?

If material questions cannot be independently answered, the conclusion must be qualified accordingly.

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21. ASSURANCE CONCLUSION MODEL

A — DEMONSTRATED

Required authority, enforcement, execution-path coverage, evidence, and reconstruction are demonstrated within the defined boundary.

B — QUALIFIED

The principal assurance condition is demonstrated, but material limitations exist.

C — NOT DEMONSTRATED

Evidence is insufficient to support the claimed assurance conclusion.

D — FAILED

Testing demonstrates violation of a required assurance condition.

Governing Rule

The assurance classification cannot be stronger than the evidence and coverage supporting it.

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22. V3.0 ASSURANCE MATRIX

Dimension	Assurance Question	Required Demonstration
Intent	What was intended?	Defined consequential objective
Authority	What authorized it?	Attributable, applicable authority
Enforcement	What controlled it?	Demonstrable enforcement
Testing	Can the claim be falsified?	Adversarial testing
Traceability	Can it be reconstructed?	Independent evidence lineage
Forensics	Can exposure/responsibility be established?	Reconstructable forensic evidence

Dimension	Assurance Question	Required Demonstration
Execution-Path Coverage	Which paths can produce consequence?	Demonstrable path coverage
Boundary Integrity	Where does the conclusion stop?	Explicit assurance boundary
External Requirements	What requirements apply?	Separate applicability/conformity assessment

23. V3.0 CORE PRINCIPLES

1. **Endpoint integrity does not prove transition integrity.**
2. **Observation does not establish authority.**
3. **Sealing does not establish enforcement.**
4. **Authorization does not prove execution.**
5. **Execution does not prove effect.**
6. **Evidence validity does not prove evidence completeness.**
7. **Captured execution does not automatically represent the complete execution population.**
8. **Propagation latency is not automatically control failure.**
9. **Propagation completion cannot retroactively authorize an earlier act.**
10. **T_5 cannot retroactively cure an invalid T_4 transition.**
11. **Delegation cannot silently expand authority.**
12. **Consequential execution paths must be covered or explicitly bounded.**
13. **Unknown coverage is an assurance limitation.**
14. **No captured act is not proof that no act occurred.**
15. **Assurance cannot exceed evidence.**
16. **Assurance cannot exceed demonstrated execution-path coverage.**
17. **The framework must remain falsifiable.**

18. **Independent reconstruction is stronger than retrospective assertion.**

19. **Every material exclusion must be explicit.**

20. **Every material coverage failure must have a predetermined assurance consequence.**

21. **Framework alignment does not establish external conformity.**

22. **Framework application does not constitute certification.**

23. **An assurance conclusion does not constitute regulatory approval.**

24. **External requirements must be assessed independently where applicable.**

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24. THE V3.0 MASTER ASSURANCE QUESTION

What independently verifiable evidence establishes that every consequential execution path within the claimed assurance population is subject to the required authorization, enforcement, testing, traceability, and forensic controls—and that the resulting assurance conclusion does not extend beyond the execution paths, evidence population, applicable requirements, and boundary that can actually be demonstrated?

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25. FINAL ASSURANCE POSITION

The Agentic AI $T_4 \rightarrow T_5$ Assurance Cone Framework™ V3.0 does not attempt to prove an impossible universal negative.

It establishes a disciplined assurance boundary.

T_4 establishes:

What the actor observed and sealed.

Act-specific evidence establishes:

What was bound to the consequential commit.

T_5 establishes:

What the authoritative enforcement environment established or enforced.

Execution-Path Coverage establishes:

Which consequential execution population the assurance claim can legitimately address.

Boundary Integrity establishes:

Where the assurance conclusion must stop.

External Requirements establish:

Which additional obligations must be assessed independently.

Therefore:

ASSURANCE ≠ ASSERTION

ASSURANCE = DEMONSTRATED CONTROL + DEMONSTRATED COVERAGE + INDEPENDENTLY VERIFIABLE EVIDENCE + EXPLICIT BOUNDARY

THE GOVERNING V3.0 RULE

IF THE EVIDENCE BOUNDARY IS NARROWER THAN THE CONSEQUENTIAL EXECUTION BOUNDARY, THE ASSURANCE CONCLUSION MUST BE NARROWER TOO.

THE CLAIMS DISCIPLINE RULE

NO ASSURANCE CLAIM MAY EXTEND BEYOND THE AUTHORITY, ENFORCEMENT, EXECUTION-PATH COVERAGE, EVIDENCE, AND BOUNDARY THAT CAN BE INDEPENDENTLY DEMONSTRATED.

THE INDEPENDENCE RULE

V3.0 MAY COMPLEMENT OTHER REQUIREMENTS, BUT IT DOES NOT SUBSTITUTE FOR THEM, AND ITS APPLICATION DOES NOT BY ITSELF ESTABLISH EXTERNAL CONFORMITY, CERTIFICATION, OR REGULATORY COMPLIANCE.

This preserves the framework as an independent, technology-neutral assurance architecture without attaching it to, naming, or implying endorsement by any external institution or standard.